

CHAPTER 3

OVERVIEW OF THE STRENGTHS AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS

3.0 INTRODUCTION

The purpose of vibration instrumentation is to accurately measure vibration amplitudes, frequencies, and phase so that a reliable determination of a machine's condition can be diagnosed. There are a whole array of vibration instruments having a wide range of capabilities. Basically, there are 5 different types of vibration measurement instruments:

- 1) Overall Level Vibration Meters (Hand-Held)
- 2) Swept-Filter Analyzers
- 3) FFT Data Collectors (also Hand-Held)
- 4) Real-Time Spectrum Analyzers
- 5) Instrument Quality Tape Recorders (both older Analog Types and newer Digital Types)

Each of these instrument types has its own optimum application in a vibration monitoring/analyzing program. When deciding which type of instrument to purchase, or use in a particular vibration test, many characteristics of the instrument must be considered. The purpose of this chapter is to give the reader a basic understanding of these characteristics and to tabulate which instrument possesses which characteristics. Table I lists many of the items which must be considered when detecting which instrument to use or which to purchase. It is important to note that the following information is meant only to reflect the general properties and capabilities of instrument configurations. Not all makes and models of vibration measurement instruments will exhibit these characteristics. However, this section provides a good checklist to review with an instrument manufacturer to fully understand the instrument's capabilities.

3.1 INSTRUMENT COMPARISONS

In order to help the reader better understand vibration measurement instrumentation, a list of common capabilities for various instrument types will be presented in this section in a glossary type format. Table I will present each type of instrument and list the general capabilities each possesses using the comparison capabilities defined below:

A) Portability - Can the equipment be easily carried around a plant or mill? How much does it weigh?

B) Typical Frequency Range - Describes in what frequency range the instrument can normally measure (usually +/-10%; or +/-3dB which equals +41%, -30%). Outside this quoted range, the instrument will normally fall off in response, and sometimes quite sharply. This will determine if the instrument has the capability to analyze the machinery in the plant based on the frequencies

generated by sources within it. Particularly watch out for either very low-speed equipment or machines having very high-speed components. Low frequency sources might include machinery with operating speeds below 100 RPM which might have hunting tooth gear frequencies, rolling element bearing cage frequencies, subharmonics far below running speed, etc. High frequencies might include gear mesh frequencies (and their harmonics), blade pass frequencies, rotor bar passing frequencies, etc. Both the analyzer and the transducer must be capable of accurately measuring and clearly displaying each of these frequencies on the machinery to be evaluated.

C) Data Measurement Format - Describes which parameter(s) each instrument type normally can measure and display for the user as follows:

- 1) **Overall Level (OL)** - Only the overall vibration amplitude is reported which cannot be broken down by its frequency components.
- 2) **Swept Filter (SF)** - A constant percentage filter (typically 2 to 10%) is swept throughout the frequency range of the instrument (typically from about 60 to 600,000 CPM in most of these instruments).
- 3) **Frequency Spectrum (FS)** - Displays amplitude versus frequency for each of the peaks in the spectrum.
- 4) **Time Waveform (TWF)** - Displays the time (seconds or minutes) versus vibration amplitude (mils, in/sec, or g).

D) Typical Display Types - Describes the type of display used by the instrument screen. Basically, there are three different types:

- 1) **Liquid Crystal Display (LCD)** Good display which is usually fairly readable under good lighting conditions depending on the type of LCD display used as well as the number of horizontal and vertical pixels used in the display. Currently, supertwist displays are probably the best LCD performer.
- 2) **Monochrome Screen (MS)** - Typically a black or green background screen with text and figures plotted in a single color (usually white, green or orange).
- 3) **Analog Meter (AM)** - Vibration amplitudes printed on a fixed background with an oscillating analog needle reporting the actual amplitude being measured.

E) Typical Transducer Types - Type of vibration pickup which generally is attached to the instrument for detecting the vibration of the equipment. See Chapter 3 for a detailed explanation of these vibration pick-ups.

- 1) **Accelerometer (A)** - Measures acceleration (g, in/sec², or mm/sec²)
- 2) **Velocity Transducer (V)** - Measures velocity (in/sec or mm/sec)
- 3) **Proximity Eddy Current Probe (P)** - Measures shaft displacement (mils or microns)

F) Phototach and/or Strobe Light Capabilities - The capability to read the input from a phototach or strobe light to determine actual 1x RPM or for use in measuring phase. Sometimes, phase might also be desired at higher operating speed multiples (2x RPM, 3x RPM, etc.)

G) Multi-Channel Availability - The capability to attach and make measurements with more than one vibration pickup at a time.

**TABLE 1. TYPICAL VIBRATION MEASUREMENT
INSTRUMENT CHARACTERISTICS @ 6/97**

CHARACTERISTICS	OVERALL LEVEL METERS	SWEPT- FILTER ANALYZERS	FFT DATA COLLECTORS	REAL- TIME ANALYZERS	TAPE RECORDERS
A. PORTABILITY Approximate Weight (Lb)	YES 1-2	SEMI 25-45	YES 5-10	SEMI 25-60	SEMI 20-40
B. TYPICAL FREQUENCY RANGE (CPM) Low End High End	120 600,000	80 600,000	60 2,400,000	0 6,000,000	VARIES VARIES
C. DATA MEASUREMENT FORMAT	OL	OL, SF	OL, FS, TWF	OL, FS, TWF	-
D. TYPICAL DISPLAY TYPES	AM	AM	LCD	MS TYPICALLY	-
E. TYPICAL TRANSDUCER TYPES	PER MANUF	A, V	A, V, P	A, V, P	A, V, P
F. PHOTOTACH AND/OR STROBE LIGHT	NO	YES	MOST USE BOTH	PTACH-YES STROBE-NO	PTACH-YES STROBE-NO
G. MULTI-CHANNEL AVAILABILITY	NO	LIMITED	SOME	YES	YES
H. SPIKE ENERGY, HFD, OR SHOCK PULSE	SOME	SOME	MOST	NO	NO
I. HIGH FREQUENCY ENVELOPED SPECTRUM	NO	NO	MOST	FEW	NO
J. SPECTRAL DISPLAY UPDATE	-	-	LT	RT	-
K. EASE OF USE	SIMPLE	SIMPLE	FAIRLY SIMPLE	DEMANDING	VARIES
L. TIME WAVEFORM STORAGE CAPABILITY	NO	HC ONLY	MOST	YES	YES
M. FREQUENCY SPECTRA STORAGE	NO	NO	YES	YES	-
N. PREDICTIVE MAINT. SOFTWARE AVAILABILITY	NO	NO	YES	SOME	-
O. NATURAL FREQUENCY TESTING CAPABILITY	NO	LIMITED	GOOD	SUPERIOR	-
P. OPERATING DEFLECTION SHAPE (ODS)	NO	NO	YES	SOME	-
Q. EXPERIMENTAL MODAL ANALYSIS CAPABILITY	NO	NO	FEW	MOST	-
R. SYNCHRONOUS TIME AVERAGING (STA)	NO	NO	SOME	YES	-
S. WATERFALL PLOTTING CAPABILITY	NO	NO	FEW	MOST	-
T. RELATIVE COSTS Low End High End Nominal Expected Cost	\$400 \$2,800 \$1,200	\$1,000 \$9,000 \$6,000	\$3,500 \$18,000 \$9,000	\$4,000 \$40,000 \$15,000	\$5,000 \$100,000 \$15,000
ABBREVIATIONS A - ACCELEROMETER AM - ANALOG METER FS - FREQUENCY SPECTRA HC ONLY - PRINTED HARD COPY ONLY LCD - LIQUID CRYSTAL DISPLAY LT - LIVE TIME (TYPICALLY UPDATED EVERY 1-4 SECONDS) V - VELOCITY TRANSDUCER MS - MONOCHROME DISPLAY OL - OVERALL LEVEL P - PROXIMITY PROBE RT - REAL-TIME (HIGH SPEED SCREEN REFRESH RATES) SF - SWEPT FILTER TWF - TIME WAVEFORM NOTE: THE ABOVE TABLE HAS NOT TAKEN INTO ACCOUNT ALL OF THE VARIOUS INSTRUMENTS AVAILABLE ON THE MARKET TODAY. DUE TO THE RAPID ADVANCES IN TECHNOLOGY, MANY PIECES OF EQUIPMENT CANNOT BE CLEARLY DEFINED INTO THE ABOVE CATEGORIES. THIS TABLE IS ONLY MEANT TO BE USED AS AN EXAMPLE OF THE CAPABILITIES A USER WOULD EXPECT TO FIND FROM A PARTICULAR INSTRUMENT TYPE. THE ABOVE CHARACTERISTICS SHOULD BE DISCUSSED IN DETAIL WITH A PARTICULAR MANUFACTURER BEFORE PURCHASING. NOTE: "CHARACTERISTIC" IS DEFINED FOR EACH LISTED ITEM IN SECTION 3.1. DISCUSSIONS COVERING THE GENERAL CAPABILITIES AND LIMITATIONS OF EACH INSTRUMENT TYPE FOLLOW IN SECTION 3.2.					

H) Spike Energy, HFD, or Shock Pulse (SPM) Measurement Capability - Ability to provide the overall level of Spike Energy, HFD (High Frequency Detected acceleration within a specified frequency range), or Shock Pulse.

I) High Frequency Enveloped Spectrum Measurement Capability - Ability to provide an enveloped spectrum of spike energy, acceleration, etc. These are referred to by vendors as a "Spike Energy Spectrum¹", "Amplitude Demodulated Spectrum²", or "Acceleration Enveloped Spectrum³", which normally are often measured within the 500 - 50,000 Hz frequency range (30,000 - 3,000,000 CPM). Related to these is the "SEE" spectrum developed by SKF Condition Monitoring, but this parameter is measured in the 250,000 - 350,000 Hz range (15,000,000 - 21,000,000 CPM).

J) Spectral Display Update - How fast does the screen refresh itself with up-to-date data. Sometimes the faster update speed may help in analysis problems, more clearly showing the problem and causing less frustration, particularly when performing natural frequency tests, or when examining electrical and/or beat frequency vibration problems.

Live Time (LT) - Screen updates approximately every 1-4 seconds depending on the particular model and settings of the instrument such as F_{MAX} , lines of resolution, and overlap processing percentage.

Real Time (RT) - Screen updates almost instantaneously, particularly in higher frequency spans. Real-Time Analyzers normally allow the lowest possible F_{MAX} setting that will still allow real-time screen updates (screen refresh rate still depends on setup parameters in an RTA just as it does in a data collector).

K) Ease of Use - An assessment of how long it usually requires to learn how to effectively operate the instrument; how much training is required; how meaningful the help screens are; and about how long it requires an analyst to become proficient at data collection or analysis with it. Also, an assessment of how demanding it is and if additional training is usually required if the instrument is only used every few weeks or months.

L) Time Waveform Storage Capability - The capability to store the time waveform signals received from the transducer for later analysis.

M) Frequency Spectra Storage - The capability to store the frequency spectra required for later analysis.

N) Predictive Maintenance (PMP) Software Compatible - Is there software available for trending and setting overall and spectral alarms for machinery in a predictive maintenance program as well as when setting up measurement routes?

O) Natural Frequency Testing Capability - The capability to conduct natural frequency tests with the instrument which may include "bump" or "impulse" tests, coastdown/runup tests, Bode or Polar plot measurement (amplitude vs. frequency vs. phase), etc.

P) Operating Deflection Shape (ODS) Capability - Ability to simultaneously measure phase and amplitude at a particular frequency and "download" this information into a personal computer outfitted with the proper software to produce operating deflected shape plots using animated plots on a computer monitor. This will actually show the machine in motion at a certain frequency such as 1x RPM (or at higher running speed multiples if the user desires and if his instrument can do so at these higher frequencies).

Q) Experimental Modal Analysis Capability - Describes the typical capability of each instrument type to measure natural frequencies, mode shapes and transfer functions. This

can often lead to a rather quick, inexpensive solution such as advising the user to weld a brace with just the right geometry and mass to resolve a natural frequency problem. Modal Analysis requires at least a two channel analyzer and modal software.

R) Synchronous Time Averaging (STA) Capability - The capability to perform Synchronous Time Averaging with the end result being that all frequencies which are not exact multiples of a designated frequency are eliminated. These nonsynchronous frequencies include running speed multiples of shafts operating at a different RPM, 2x Line Frequency, bearing defect frequencies, belt defect frequency or speed, etc. Note that these nonsynchronous frequencies will disappear from both the spectrum and time waveform if a sufficient number of averages are taken (often 250 to 500 averages). This can be of great assistance to the analyst attempting to evaluate a machine with significant vibration transmitted into it from its neighboring machines from a felt on a paper machine, or even from nonsynchronous frequency components within the machine rotor itself being evaluated since again it will limit the vibration spectrum being measured only to multiples of the frequency which are synchronous with the trigger source (such as a phototach or strobe light).

S) Waterfall or Cascade Plotting Capability - The capability to display either data collected during a "run-up" or "coastdown"; or data collected from one PMP survey to the next directly on the spectral display of the instrument in question.

T) Relative Costs - An estimate of the current costs for these types of machines. These ranges may vary considerably, based on the "extras" such as software, cabling, number of channels, and auxiliary equipment which may be purchased. Please note the "Nominal Cost" which is what one should expect to pay for most of the instruments of each type, and outfitted with the usual number of "bells and whistles" for an instrument in this class.

3.2 GENERAL CAPABILITIES OF EACH VIBRATION INSTRUMENT TYPE

The following is a summary of the instruments covered previously in Table I outlining both their major advantages and drawbacks.

3.21 OVERALL LEVEL VIBRATION METERS

Overall level vibration meters are used to measure so-called total vibration in the form of acceleration, velocity, or displacement which is then most often displayed on an analog meter (many of these meters also measure one of the high frequency banded parameters such as spike energy (gSE)). They are lightweight and very portable which makes them ideal for collecting data from in-service machinery around a plant or mill. These instruments have been around for years and, at one time, were the main instruments used to evaluate large quantities of machines in typical predictive maintenance programs of yesteryear (the analyst recorded his measurements on a clipboard taken on the route). Plots of overall vibration versus time (trend plots) were then manually generated using these results.

However, the overall level vibration meter is not now a very useful tool in a vibration monitoring/analysis program due to its lack of capability. They are only possibly useful in monitoring noncritical machinery since the vibration level at specific frequencies cannot be measured by this instrument. In this case, potentially severe problems can develop which have negligible effect on the overall level (i.e., bearing wear, gear wear, cracked gear teeth, cracked rotor bars, etc.)

The overall vibration meter does not have the capability to measure vibration amplitude versus frequency content (frequency spectrum).

Frequency content is very important since even very low levels of vibration at certain frequencies can signify definite machine problems. Most of these instruments likewise do not have the capability of storing the measured vibration signal or frequency output. The vibration reading must be manually recorded creating a cumbersome data collection procedure. Also, the frequency range of the instrument may be limited and should be checked before use. Furthermore, it is not capable of displaying the time-based waveform (amplitude vs. time). Worst of all, numerous failures can occur with only a minute increase (or decrease) in the overall level, particularly dependent on which vibration parameter is being measured and trended (displacement, velocity, or acceleration).

Finally, even in the case of overall level meters which can measure the high frequency banded acceleration (such as overall spike energy), these instruments can even be rendered somewhat ineffective. For example, if the user feels he can monitor bearing health simply by monitoring overall spike energy, HFD, or shock pulse, he should be aware that each of these are effected not only by rolling element bearing condition, but also by lubrication, cavitation, high pressure steam or air, gear condition, rotor rub and belt squeal. To determine what is causing this parameter to increase will require spectrum analysis (hopefully using this parameter vs. frequency).

3.22 SWEPT-FILTER ANALYZERS

Swept filter analyzers are normally an older technology instrument which are an enhancement over the overall level vibration meter alone by providing a means for extracting the filtered vibration level at various frequencies. It is used by adjusting a tuning knob which filters out any vibration signals outside the filter frequency. For example, using a 10% swept filter, measuring a frequency at about 1000 CPM would actually include the amplitude within the 950 to 1050 CPM range. Only the vibration level for the frequency range being swept is reported on its analog meter display. Typically, the filters employed on these units vary from approximately 2% to 10%, meaning that frequency precision is limited (and, the higher the frequency, the lower the precision of the frequency reading in terms of CPM or Hz).

This type of equipment has nearly been replaced by the capabilities of the newer FFT data collectors with higher precision frequency spectrum and also data storage capabilities. However, they can still be of some use for field balancing, strobe light "slow motion studies", phase analysis with the use of the strobe, determining natural frequencies of components (to some extent) and determining vibration amplitude at various frequencies. Outputs to an analog X-Y plotter are available on some of these units, and even built in on others.

These instruments are typically large and heavy in size, and not very portable. This is acceptable for analyzing one machine. However, it now would be a nightmare to use them exclusively on a predictive maintenance route. The data cannot be stored for further analysis or off-loaded to a computer. The entire process of collecting frequency data is very time consuming and cumbersome. Also, they are not normally capable of displaying the time-based waveform.

3.23 FFT PROGRAMMABLE DATA COLLECTORS

FFT data collectors are the current state-of-the-art instruments of choice for predictive maintenance programs and also can be useful in a variety of analysis situations. The FFT (Fast-Fourier Transform) capability allows the time waveform captured by these units to be transformed into a frequency spectrum and displayed on a small LCD screen in live-time (typically within 1-4 seconds depending on the data collection mode, along with the chosen frequency span, and lines of resolution settings on the instrument). The control panel allows rapid switching of the display parameters which is helpful when performing monitoring on many different machines and their components, or when performing analysis on troubled

equipment. Also, increasing the speed of some data collectors is the ability to simultaneously measure 3 different parameters with one push of the "store" button (such as velocity, acceleration, and spike energy).

The data collector was designed for data collection on many pieces of machinery within one outing. It is portable and has the capability of storing data for further analysis back at the office. Typically, software designed to work with the data collector is installed on a personal computer which can download a route into or receive data from the data collector for analysis of the machine's condition. Also, the data collector can be used as an analyzer in the field due to its graphics, FFT, and live-time capabilities. Most of them are also capable of time waveform presentation. Furthermore, a phototach or strobe light can be attached to most of today's data collectors to measure phase at various points. Most of them can also measure high frequency parameters such as HFD, Spike Energy, etc., used for identification of problems such as bearing deterioration, worn gear teeth, lubrication problems, cavitation, and rubs. In addition, many of these units can now generate a spectrum of one of these high frequency parameters (i.e., spike energy spectrum^①, amplitude demodulated spectrum^②, acceleration enveloped spectrum^③, etc.)

The LCD display of some data collectors is "live-time" which typically updates on the average of once every 1 to 4 seconds as compared with a real time spectrum analyzer which has nearly instantaneous updates (again depending on frequency range). Note that some data collectors have markedly faster live-time update speeds than others. Only a few FFT data collectors are capable of multi-channel data input which is helpful in operating deflection shape tests or modal analysis, as well as for greater data collection speed.

The frequency range of the average data collector is normally limited to about 60 CPM (1 Hz) on the low end and up to about 1,500,000 CPM (25,000 Hz) on the high end which may pose some analysis or data collection problems. However, some data collectors now are available which allow measurements down to as low as approximately 6 CPM (0.10 Hz) and up to as high as 4,500,000 CPM (75,000 Hz). Real-time spectrum analyzers can sometimes measure frequencies down to almost 0 CPM, and up to about 6,000,000 CPM (0 - 100,000 Hz). With the continued intense development of data collectors taking place today, no telling what their capabilities might be within the next 5 to 10 years!

In terms of memory, one of the more major enhancements to some of the data collectors has been the addition of PCMCIA cards, which give them practically unlimited storage. Also, some of these PCMCIA cards allow for the loading of special functions into the instruments such as balancing, transient capture, natural frequency testing, etc.

Notes:

- ①= Spike Energy Spectrum (Available from IRD Mechanalysis, based in Columbus, OH)
- ②= Amplitude Demodulated Spectrum (Available from CSI, based in Knoxville, TN)
- ③= Acceleration Enveloped Spectrum (Available from SKF Condition Monitoring based in San Diego, CA)

3.24 REAL-TIME SPECTRUM ANALYZERS

The real-time spectrum analyzer is the most powerful diagnostic tool on the market for advanced diagnostic techniques. There are a variety of them now available having a wide range of capabilities. In layman's terms what is meant by a "real-time" analyzer is that the user is usually able to simultaneously see the oscillations of all peaks within the frequency span chosen. That is, if one had a 2-pole, 3590 RPM motor belt driving a 1000 RPM fan in close proximity to another fan operating at about 1050 RPM, the user should be able to see two sets of pulsating

frequencies - one set would be the two fan speeds of 1000 and 1050 RPM which would likely set up a "beat frequency" of 50 CPM thereby causing the amplitudes of both the 1000 and the 1050 RPM peaks to pulsate up and down; the second set of pulsating frequencies would likely be located at 2x RPM of the motor ($2 \times 3590 = 7180$ CPM) and the frequency at 2x electrical line frequency ($2 \times 3600 = 7200$ CPM). If one set up the proper frequency range and number of FFT lines, he should readily see this other peak beating between 7180 and 7200 CPM frequencies.

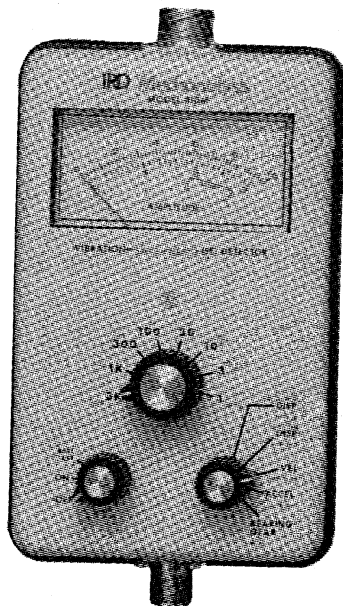
The typical real-time analyzer is capable of measuring frequencies in a wide range from about 0 to 6,000,000 CPM (0 - 100,000 Hz) which is important for diagnosing gear problems and blade pass frequency problems on high-speed machines (however, the limitations here will most always be the frequency range capabilities of the transducer employed, as well as the method used to mount the transducer itself). Multi-channel capabilities, available in many of these units, allow excellent data capture for operating deflection shape and modal analysis, as well as speeding up the process of data collection. Most all real-time analyzers are also typically capable of performing synchronous time averaging (STA), order tracking, and are able to perform a wide variety of natural frequency tests.

The real-time display can be very helpful in diagnosing problems as opposed to the live-time display in data collectors. The screen updates "faster than the eye" in a real-time analyzer when the F_{MAX} is properly specified. Phototach input is available for measuring and comparing phase between measured points and for many different frequencies simultaneously. Real-time analyzers are excellent instruments for performing impulse natural frequency tests due to their "peak hold" capabilities. Furthermore, "peak hold" can be used to effectively monitor coastdown and run-up tests when attempting to identify system natural frequencies. Also, Bode, Polar and Waterfall plots can be generated using real-time analyzers as another verification of natural frequencies (as well as identifying the component(s) which is resonant). These analyzers can likewise be used in a fashion similar to oscilloscopes to generate shaft orbit plots, computerized order tracked spectra, etc. Some of these analyzers are also outfitted with digital time buffers which can be used in a manner similar to tape recorders, capturing transient events, runups, coastdowns, etc. from data captured and stored to hard disk in the field (giving such data file names so they can be evaluated and reported upon at convenience by the analyst).

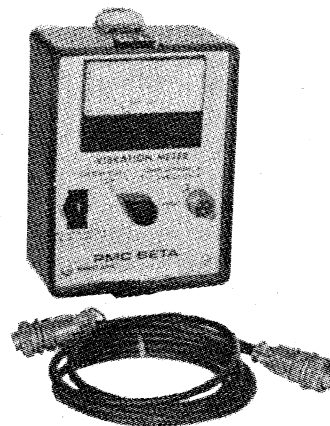
The real-time spectrum analyzer is a more complex piece of equipment which requires additional training and frequent use to remain proficient in its use. It is not very portable and should only be used for machinery troubleshooting and specialty monitoring. Memory is typically limited in comparison with the PMP data collectors. However, some of them are now outfitted with 3.5 inch (1.44 Mb) floppy drives which gives them almost unlimited spectral and time waveform storage. Also, some real-time analyzers are now composed of a computer outfitted with special cards, making them an even more useful instrument and providing them with memory limited only by hard disk size. In these cases, it is sometimes possible to install PMP software on some of these units thereby allowing their spectra to be passed back into the PMP database. Finally, some of these computerized real-time analyzers allow use of word processing, spreadsheets and graphics software thereby allowing "on-the-spot" report generation if desired.

3.25 INSTRUMENT QUALITY TAPE RECORDERS

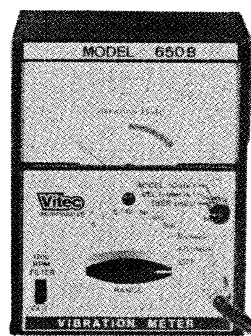
Instrumentation quality tape recorders are useful devices for machinery troubleshooting and for simultaneously recording many different signals (whether from vibration pickups, or from pressure transducers, tachometer signals, current transformers, phototachs, etc.). Both analog and digital tape recorders are available. The analog type records the actual signal input from the transducer without breaking it into a number of sampled points similar to a voice recorder. However, analog units today are usually limited to a dynamic range of only about 40 to 48 dB. The digital recorder samples the input at a specified rate and reproduces it as stored numbers with the advantage of a much larger dynamic range capability (now on the order of about 78 dB). That



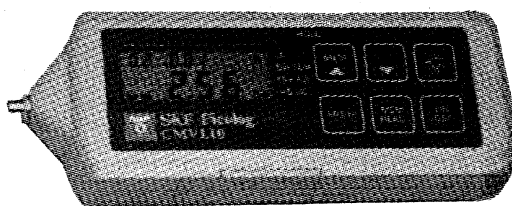
**A. IRD 810 VIBRATION SPIKE
ENERGY DETECTOR**



**B. PMC/BETA 201A
VIBRATION METER**



**C. VITEC 650
VIBRATION METER**

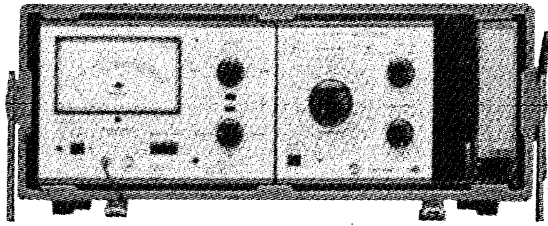


**D. SKF CMVLIO PICOLOG
VIBRATION / SEE / ACCELERATION
ENVELOPE DETECTOR**

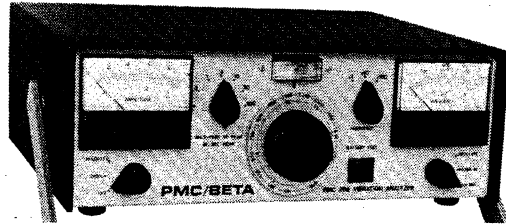


**E. SKF CMVP20
VIBRATION PEN**

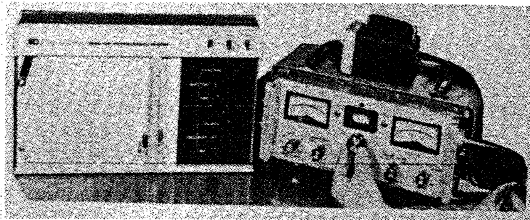
FIGURE 1. EXAMPLES OF VARIOUS HAND-HELD OVERALL VIBRATION METERS



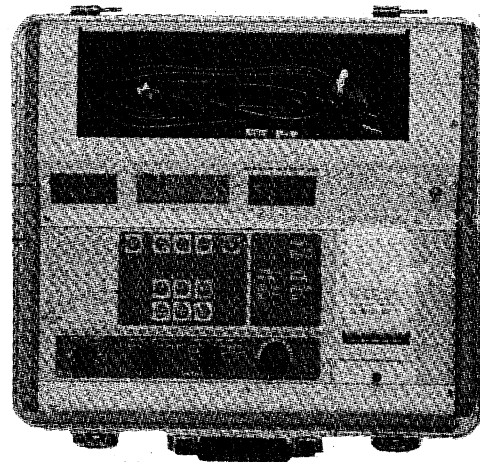
A. B&K 3513



B. PMC/BETA 208



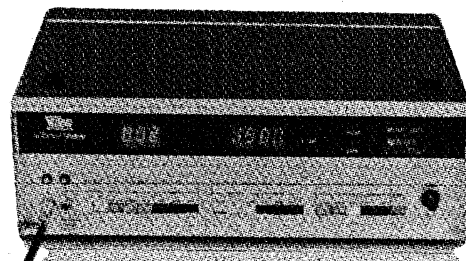
C. IRD 350 WITH
IRD 1080 PLOTTER



D. IRD 880



E. SCHENCK VIBRATION 30

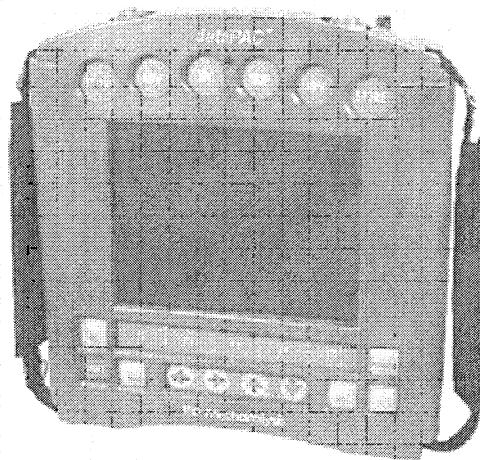


F. VITEC VIBRA/VIEW

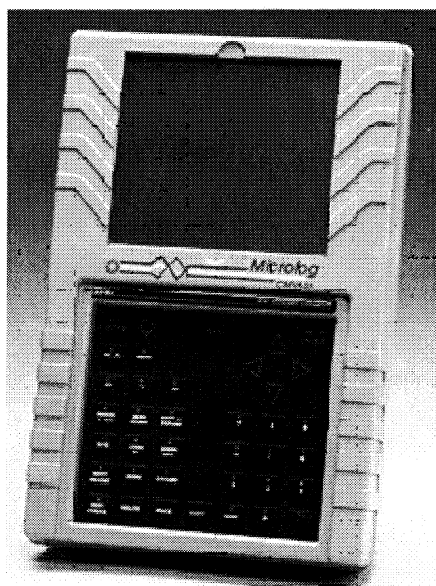
FIGURE 2. EXAMPLES OF PORTABLE SWEPT-FILTER ANALYZERS



**A. CSI 2120
MACHINERY ANALYZER**



ENTEK IRD DataPac 1500



C. SKF MICROLOG CMVA55

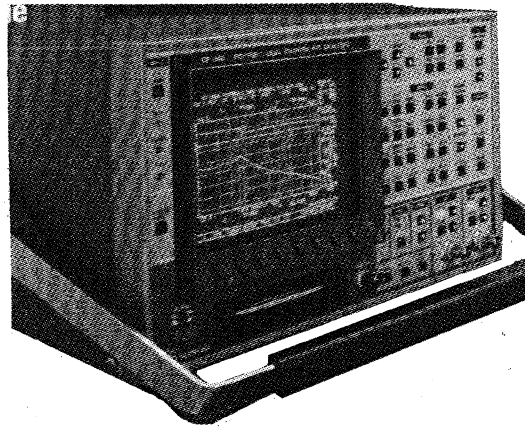


**D. PREDICT/DLI WATCHMAN DC-7B
(3 CHANNEL)**

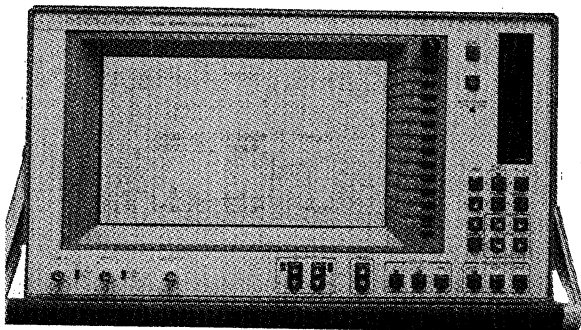
FIGURE 3. EXAMPLES OF VARIOUS DATA COLLECTORS



**A. HEWLETT PACKARD 35670A
DYNAMIC SIGNAL ANALYZER
(2 OR 4 CHANNEL)**



**B. ONO SOKKI CF-360
(2 CHANNEL)**



**C. ROCKLAND 90665L
DYNAMIC SIGNAL ANALYZER
(2 CHANNEL)**

**D. SIGNAL PROCESSING SYSTEMS SA 390
DYNAMIC SIGNAL ANALYZER
(2, 4, 6, OR 8 CHANNEL)**

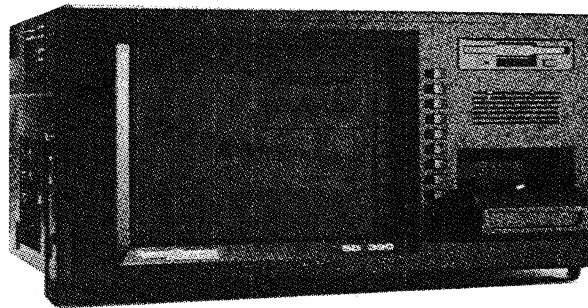
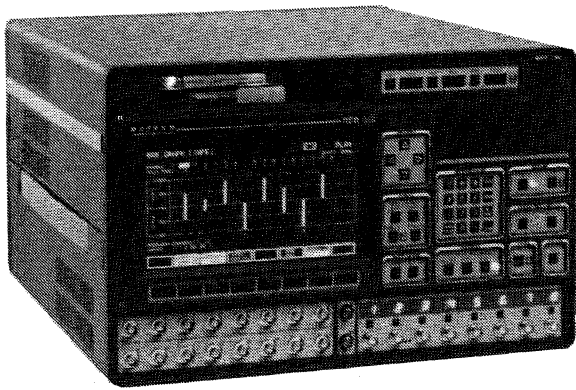


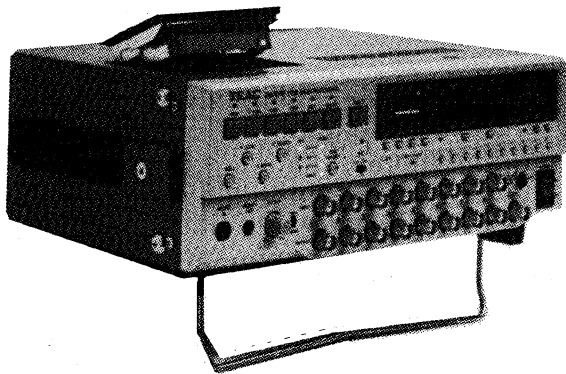
FIGURE 4. EXAMPLES OF VARIOUS REAL-TIME SPECTRUM ANALYZERS



**A. METRUM RSR 512 DAT RECORDER
(8,16,24 OR 32 CHANNEL)**



**C. SONY PC 116 DAT RECORDER
(2 THRU 16 CHANNEL)**



**B. TEAC RD-130T DAT RECORDER
(2, 4, OR 8 CHANNEL)**



**D. SONY TCD-D10 PRO II DAT RECORDER
(2 CHANNEL)**

FIGURE 5. EXAMPLES OF INSTRUMENT QUALITY TYPE RECORDERS

is, digital tape recorders commonly have more than 30 dB higher dynamic range, which means they are more than 31 times as amplitude sensitive (and therefore can detect low amplitude frequencies such as bearing frequencies of only about 0.01 in/sec in the presence of much higher amplitude peaks such as a running speed amplitude of more than 1.0 in/sec). Such a capability would be out of the question with the older analog tape recorder models.

Tape recorders allow the analyst the capability of very accurately recording the time waveform of the vibration signal for future analysis with measurements made by a real time spectrum analyzer. In addition, they afford him the luxury of employing a wide range of frequency range spectra back at the office. Both types of tape recorders have multi-channel capabilities (up to 64 channels or more) which allow the user to capture many data points simultaneously. This is especially useful when many points need to be analyzed, or if machine testing requires that the machine be stopped resulting in lost production. Input can be brought in from a wide variety of signals such as voice or noise, pressure, tachometer signal (RPM), vibration, or electrical current from which FFT spectra can be generated and evaluated. Tape recorders will also capture short-lived transient events which cannot even be "seen" by an analyzer. After the signal has been captured on tape, the user has excellent flexibility in analyzing the signal back in the office since it may be played back over and over again looking at alternate spectra settings. For instance, frequency span can be adjusted to provide the necessary resolution. Also, anomalous events can be analyzed in many different settings to determine the source.

However, depending on the capabilities of the recorder (analog vs. digital, dynamic range, etc...), instrument quality tape recorders can be very costly. Furthermore, they may be complicated to use and could require frequent use to remain proficient in data collection.

REFERENCES

1. "Spike Energy Spectrum" - developed by IRD Mechanalysis, Inc., based in Columbus, OH.
2. "Amplitude Demodulated Spectrum" - developed by Computational Systems, Inc. (CSI), based in Knoxville, TN.
3. "Acceleration Enveloped Spectrum" - developed by SKF Condition Monitoring with USA headquarters in San Diego, CA.